

Knowledge representation

LECTURE 2

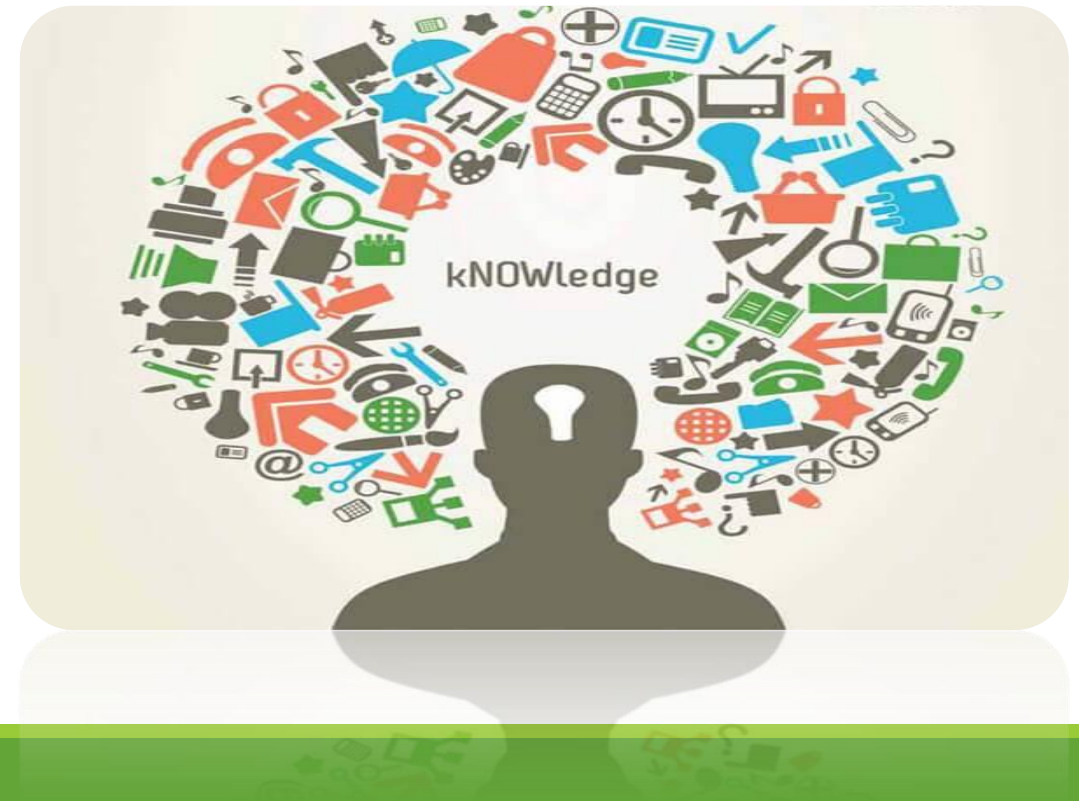
What is knowledge?

facts, information, and skills acquired through experience or education; the theoretical or practical understanding of a subject.

 Knowledge = information + rules

EXAMPLE

Doctors, managers.



What is Knowledge representation?

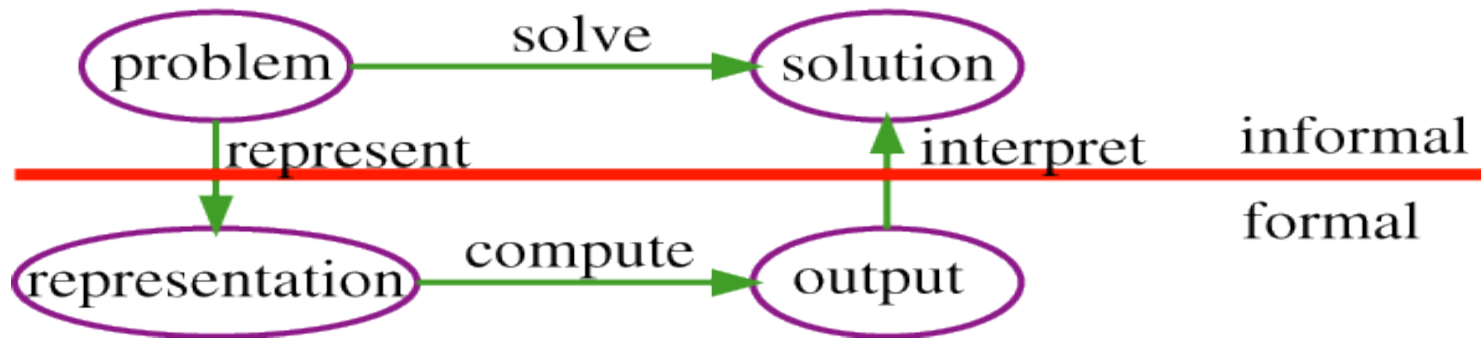
☆ Knowledge representation is a relationship between two domains.

هو ربط بين عالَمين

العالم الحقيقي (Real world)

الكمبيوتر (Machine world)

Knowledge representation (KR) is the field of **artificial intelligence (AI)** that representing information about the world in a form of computer system, that can solve complex tasks, such as diagnosing a medical condition.



مكان فيه كل المعرفة الخاصة بمجال معين

Quickly what is knowledge base.

A knowledge base comprises a collection of facts, rules, and relationships about a specific domain, represented in a formal language that a computer system can process. It usually includes:

1. **Meta structural Database:** Defines the structure and organization of the knowledge.
2. **Database Semantics:** Specifies the meaning of the data and relationships.
3. **Primary Information:** Contains the core facts and data ذات الصلة pertinent to the domain.
4. **Primary Methods of Data Processing:** Encompasses the procedures and rules for manipulating and reasoning with the data.

القواعد + العمليات

:للي نستخدمها عشان نفكر ونستنتج زي

IF-THEN rules

There are 5 types of knowledge according to the nature and function of the knowledge

There are 5 types of knowledge.

- 1) Procedural k.
- 2) Declarative k.
- 3) Meta k.
- 4) Heuristic k.
- 5) Structural k.

HOW = Procedural

1) Procedural Knowledge

Gives information/ knowledge about how to achieve something.

Describes how to do things provides set of directions of how to perform certain tasks.

Procedural knowledge, also known as imperative/الزامي knowledge, is the **knowledge** exercised in the performance of some task.

Example

How to drive a car?

FACTS = Declarative

2)Declarative knowledge

Declarative knowledge, also referred to as verbal or factual knowledge, describes things, events, or processes; their attributes; and their relation to each other.

هو عكس البروسيدجول
It is contrary to procedural or implicit knowledge, which refers to the knowledge of how to perform or operate.

"Can this knowledge be true or false?"

Example

It is sunny today

I ate oatmeal for breakfast this morning'

META = knowledge about knowledge
and how to gain them

3)Meta Knowledge

It's a knowledge about knowledge and how to gain them.

Example

— The knowledge that blood pressure is more important for diagnosing a medical condition than eyes color.

Heuristic = Experience / Tricks
rules f thumb/ shallow knowledge

4)Heuristic Knowledge

Representing knowledge of some expert in a field or subject.

Rules of thumb, Heuristic Knowledge are sometimes called shallow knowledge.

Heuristic knowledge are empirical/ تجريبية as opposed to deterministic.

Example:

If you're traveling to a popular destination, book flights and hotels several months in advance to save money

Structure = Relationships

5)Structural Knowledge

Describes what relationship exists between concepts/ objects. Describe structure and their relationship.

Example1:

How to various part of car fit together to make a car, or knowledge structures in term of concepts, sub concepts and objects.

Example2:

University Structure (Simple & Clear)

University
→ contains → Faculties

Faculty
→ contains → Departments

Department
→ offers → Courses

Classification according to **source**, **nature**, and **method of representation** or **use**.

Knowledge Type	Description
Domain knowledge	Domain knowledge is valid knowledge for a <u>specified domain</u> . Specialists and experts develop their own domain knowledge and use it for problem solving. دكتور → طب
Meta knowledge	Meta knowledge can be defined as knowledge about knowledge.
Commonsense knowledge	Common sense knowledge is a general purpose knowledge expected to be <u>present in every normal human being</u> . Common-sense ideas tend to relate to events <u>within human experience</u> .
Heuristic knowledge	Heuristic is a specific rule-of-thumb or argument <u>derived from experience</u> .
Explicit knowledge	Explicit knowledge can be easily <u>expressed in words/numbers</u> and shared in the form of data, scientific formulae, <u>product specifications</u> , manuals, and universal principles. It is <u>more formal</u> and systematic. معرفة مكتوبة وسهلة التوثيق
Tacit knowledge	Tacit knowledge is the knowledge stored in <u>subconscious mind</u> of experts and <u>not easy to document</u> . It is highly <u>personal</u> and hard to formalize, and hence difficult to represent formally in system. Subjective insights, <u>intuitions</u> , <u>emotions</u> , <u>mental models</u> , values and actions are examples of <u>tacit knowledge</u> . مهارة الرسم

Table 1: Types of Knowledge

For Reading

Type of Knowledge	Representation	Examples
Declarative Knowledge	Logical Representation	Mathematical rules, business logic
Conceptual/Hierarchical	Semantic Networks	Taxonomies, ontologies
Structured Knowledge	Frames	Object attributes, scenarios
Procedural Knowledge	Production Rules	Expert systems, decision rules
Domain-Specific Knowledge	Ontologies	Medical ontologies, NLP systems
Probabilistic Knowledge	Bayesian Networks	Risk assessment, diagnosis
Implicit Knowledge	Neural Networks	Image/speech recognition
Tabular Knowledge	Relational Databases	Business records, inventories
Experiential Knowledge	Case-Based Representation	Legal cases, medical diagnoses

Knowledge Representation and Reasoning

بناخذ معلومات من العالم الحقيقي
نحولها لشكل يفهمه الكمبيوتر

- ✓ Knowledge Representation and Reasoning (**KR, KRR**) represents information from the real world for a computer to understand and then utilize this knowledge to solve **complex real-life problems** like communicating with human beings in natural language.
تخزين + فهم + استخدام
- ✓ Knowledge representation in AI is not just about storing data in a database, it allows a machine to learn from that knowledge and behave intelligently like a human being.

إليه أنواع المعلومات التي لازم نخططها في السيستم؟

What kind of knowledge to Represent:

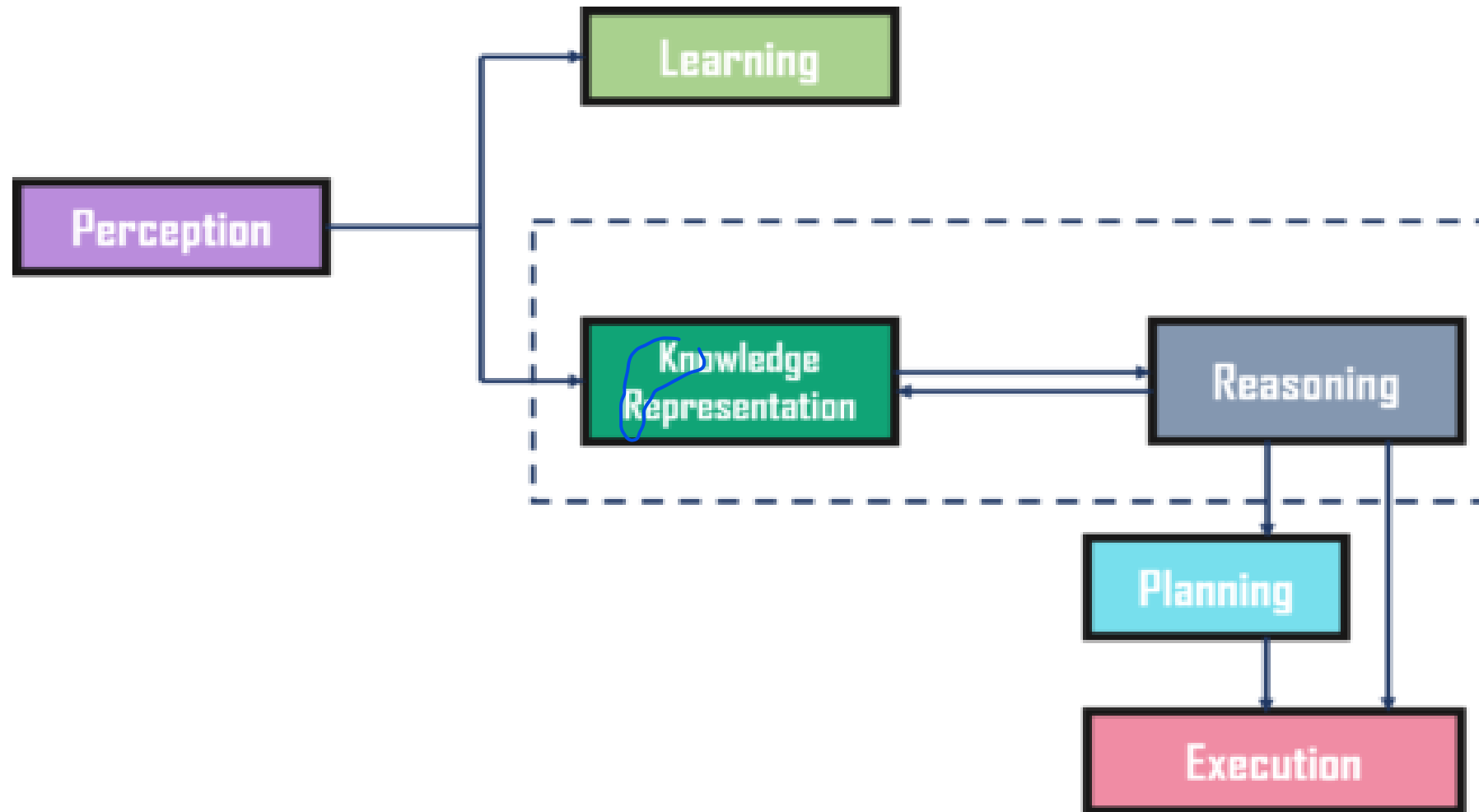
The different kinds of knowledge that need to be represented in AI include:

- ✓ **Objects** - All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.
 - ✓ **Events** - Events are the actions which occur in our world. السلوك / إزاي نعمل حاجة
 - ✓ **Performance** - It describe behavior which involves knowledge about how to do things
 - ✓ **Facts** - Facts are the truths about the real world and what we represent
 - ✓ **Meta-Knowledge** It is knowledge about what we know.
- المكان الذي كل ده بيتخزن فيه ويكن عباره عن سينتسنيز
- ✓ **Knowledge-base** - - The central component of the knowledge-based agents is the knowledge base. The Knowledge base is a group of the Sentences (Here, sentences are used as a technical term and not identical with the English language).

Cycle of Knowledge Representation in AI

Artificial Intelligent Systems usually consist of various components to display their intelligent behavior. Some of these components include:

- 1 ✓ **Perception**
- 2 ✓ **Learning**
- 3 ✓ **Knowledge Representation & Reasoning**
- 4 ✓ **Planning**
- 5 ✓ **Execution**



- Perception component

- ✓ The Perception component retrieves data or information from the environment.
- ✓ with the help of this component, you can retrieve data from the environment
- ✓ Find out the source of noises and check if the AI was damaged by anything.
- ✓ Also, it defines how to respond when any sense has been detected. كمان يحدد رد الفعل
 - ✓ A self-driving car uses cameras, radar, and LiDAR to detect pedestrians, traffic signs, and other vehicles.

2 Learning Component

- ✓ There is the Learning Component that learns from the captured data by the perception component.
- ✓ The goal is to build computers that can be taught instead of programming them.
- ✓ Learning focuses on the process of self-improvement.

It may involve:

- Acquiring new facts
- Deriving new rules
- Updating heuristics
- Improving search efficiency
- Refining decision boundaries

In symbolic systems:

- Learning may add new rules to the knowledge base.

In ML systems:

- Learning adjusts weights or internal representations.

✓ A recommendation system like Netflix learns user preferences by analyzing past viewing habits to suggest movies or TV shows.

Main Component

يحول المعرفة لشكل يفهمه الكمبيوتر

- 3 ✓ The **main component** in the cycle is **Knowledge Representation and Reasoning** which shows the human-like intelligence in the machines.
- ✓ Knowledge representation is all about understanding intelligence.
- ✓ Instead of trying to understand or build brains from the scratch, its goal is to understand and build intelligent behavior from the top-down and focus on **what** needs to be known in order to behave intelligently.
- ✓ Also, it defines how automated reasoning procedures can **make this knowledge available as needed**.
 - ✓ In a medical diagnosis system, knowledge about diseases, symptoms, and treatments is stored. Reasoning is used to match patient symptoms to possible diagnoses and suggest treatments.

└ Planning and Execution components

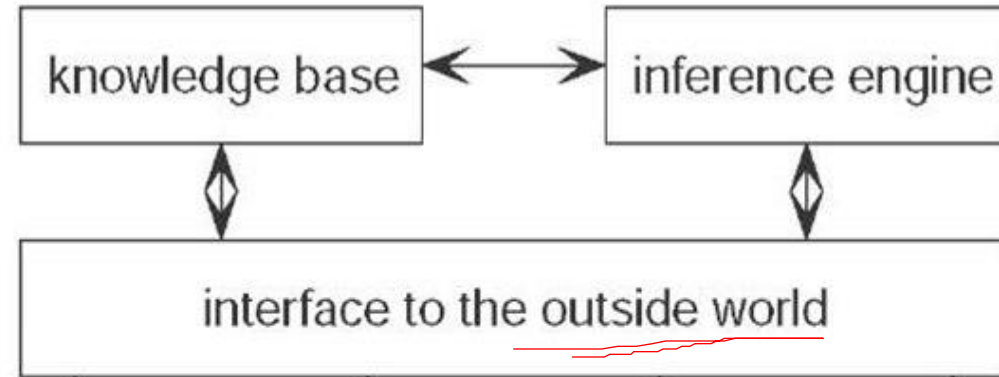
- ✓ The Planning and Execution components depend on the analysis of knowledge representation and reasoning.
- ✓ Here, planning includes giving an initial state, finding their preconditions and effects, and a sequence of actions to achieve a state in which a particular goal holds.
- ✓ Now once the planning is completed, the final stage is the execution of the entire process.
 - ✓ A robot tasked with cleaning a room plans the most efficient route to vacuum all areas while avoiding obstacles.
 - ✓ A delivery drone executes the plan by flying to the delivery location, dropping off the package, and returning to the base.

KBS Architecture

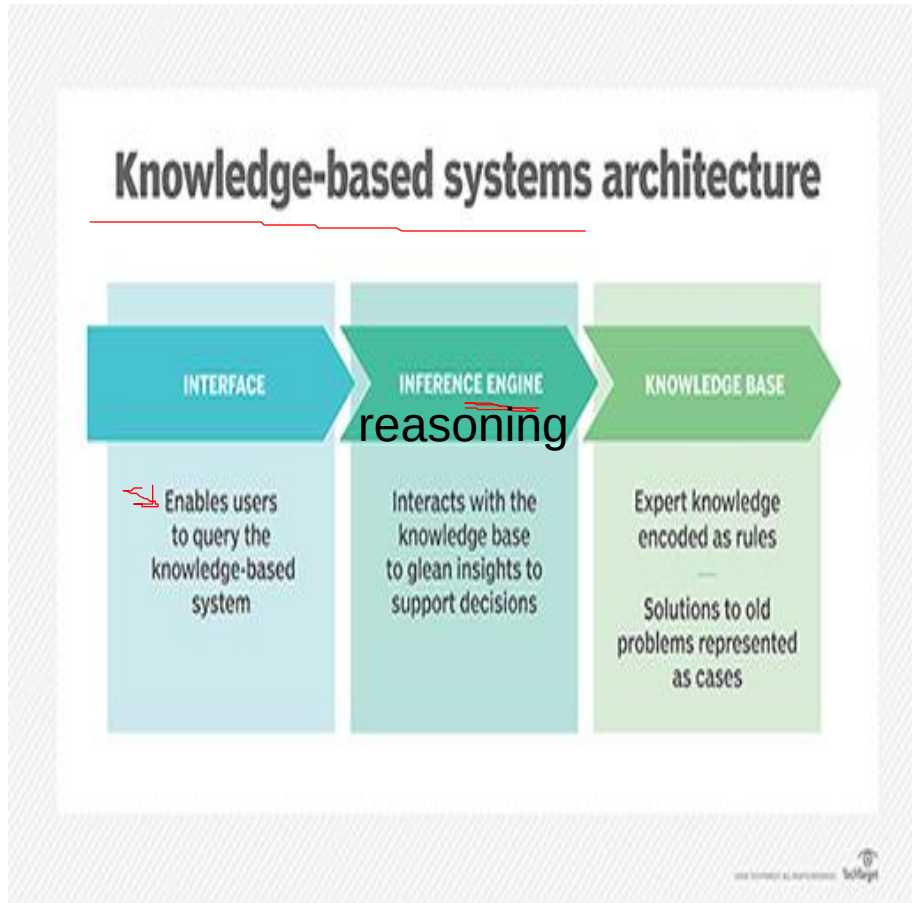
👉 KRR (Representation + Reasoning)

📌 KB = Representation

📌 Inference Engine = Reasoning



- Knowledge module is KB
- Control Module is Inference Engine



Basically 4 types of knowledge representation in AI

- 1) Logical representation
- 2) Production rule
- 3) Semantic networks
- 4) Frame representation

1) LOGICAL REPRESENTATION

Logic is based on truth. ي صح ي غلط مفيش فالنص

There are 2 types of LR

1) propositional logic(PL)

2) first order predicate logic(FOL)

Propositional Logic

Propositional logic (PL) is the simplest form of logic where all the statements are made by propositions.

A proposition is a declarative statement which is either true or false.

It is a technique of knowledge representation in logical and mathematical form.

Example: ينفع نقول صح أو غلط

- a) It is Sunday.
- b) The Sun rises from West (False proposition)
- c) $3+3=7$ (False proposition)
- d) 5 is a prime number.

مش propositions

:"Where are you?" "How are you?"

True/False دي أسئلة → مش

Some basic facts about propositional logic:

- Propositional logic is also called **Boolean logic** as it works on 0 and 1.
- In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol for representing a proposition, such A, B, C, P, Q, R, etc.
A: "It is raining."
B: "The ground is wet."
بنرمز للجمل بحروف
- Propositions can be either true or false, but it cannot be both.
- Propositional logic consists of an object, relations or function, and **logical connectives**.
Objects: Numbers
Relation: "greater than"
Let P : " $5 > 3$ " $\rightarrow$ True
- These connectives are also called logical operators. And, OR, Not



Some basic facts about propositional logic:

- The propositions and connectives are the basic elements of the propositional logic.

- P : "The exam is easy."
- Q : "Students are happy."

$$P \rightarrow Q$$

حاجت تودي الى حاجه

- $P, Q \rightarrow$ propositions
- $\rightarrow$ connective

- Connectives can be said as a logical operator which connects two sentences.

- Ali studies AND Ali passes."

- A proposition formula which is always true is called **tautology**, and it is also called a valid sentence. $P \vee \neg P$

بي صح او غلط كذا حاجه دايمًا هتحصل عشان دي كل

- A proposition formula which is always false is called **Contradiction**. $P \wedge \neg P$

الاحتمالات

بي صح و غلط عمرها مهتحصل

- Statements which are questions, commands, or opinions are not propositions such as "Where is Rohini", "How are you", "What is your name", are not propositions.

Syntax of propositional logic:

The syntax of propositional logic defines the allowable sentences for the knowledge representation. There are two types of Propositions:

1 **Atomic Propositions**

2 **Compound propositions**

Atomic Proposition: Atomic propositions are the simple propositions. It **consists of a single proposition symbol**. These are the sentences which must be either true or false. فيها حرف واحد

Example:

- a) $2+2$ is 4 , it is an atomic proposition as it is a **true** fact.
- b) "The Sun is cold" is also a proposition as it is a **false** fact.

P

Cont.

Compound proposition: Compound propositions are constructed by combining simpler or atomic propositions, using parenthesis and logical connectives.



Example:

$P \wedge Q$

- a) "It is raining today, and street is wet."
- b) "Ankit is a doctor, and his clinic is in Mumbai."

First Order Logic

- Statements that cannot be made in propositional logic but can be made in FOL.
- When you paint a block with green paint, it becomes green. – In propositional logic, one would need a statement about every single block, one cannot make the general statement about all blocks.
- "مش بيعرف يعبر عن "كل" أو "بعض"
When you sterilize a jar, all the bacteria are dead. – In FOL, we can talk about all the bacteria without naming them explicitly.
- A person is allowed access to this Web site if they have been formally authorized, or they are known to someone who has access.

First Order Logic

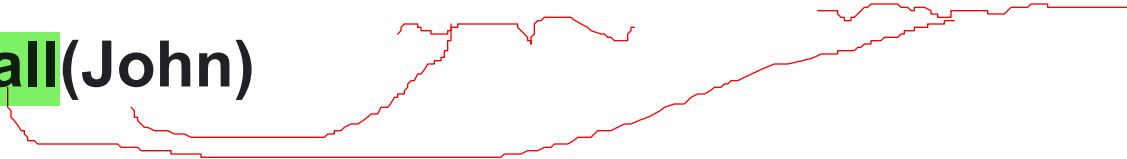
- First-order logic is another way of knowledge representation in artificial intelligence. It is an extension to propositional logic.
- **First-order logic** is also known as **Predicate logic** or First-order predicate logic.
- Propositional logic assumes the world contain **facts**.
- While a **Predicate** is used to show “A description, property, or relationship applied to objects.”

Predicate Logic Example

A sentence in **first-order logic** is written in the form Px or $P(x)$, where **P is the predicate** and x is **the subject(object)**, represented as a variable.

John is tall. **John is Subject and tall is predicate**

tall(John)



Arun and Vijay are friends.

friends(Arun , Vijay).

Basic Elements of First Order Logic

Elements	Examples
Constant	1, 2, Pune, Ajay...
Variables	x, y, a , b,...
Predicates صفات أو علاقات	Brother, Father,...
Function	Sqrt, LeftLegOf,...
Connectives	$\wedge$, $\vee$, $\neg$, (And, Or, Not)
Equality	$=$ $\longrightarrow$ $\longleftrightarrow$
Quantifier	for all there exists $\forall$, $\exists$ في واحد على الاقل

Quantifiers

In First Order Logic , a sentence can be structured using the Universal quantifier or the Existential quantifier.

Universal Quantifier

- $\forall <variable> <sentence>$
- True for all.
- We use implication " $\rightarrow$ "
(conclusion)

Existential Quantifier

- $\exists <variable> <sentence>$
- True for atleast one.
- We use conjunction symbol " $\wedge$ "
(connectors)

Some Examples of FOL

1. All kings are person.

$\forall x: \text{King}(x) \implies \text{person}(x)$

2. No-one likes noodles.

$\forall x: \text{person}(x) \implies \neg \text{likes}(x, \text{noodles})$

3. Some dog is a pet.

$\exists x: (\text{dog}(x) \wedge \text{pet}(x))$ **atleast one a dog and a pet**

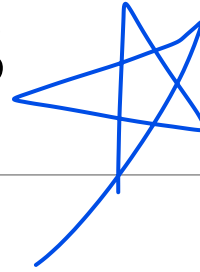
4. Every student pass or fail.

$\forall x: (\text{student}(x) \implies (\text{pass}(x) \vee \text{fail}(x)))$

5. If x is a parent of y, then x is older than y.

$\forall x \forall y \text{ Parent}(x,y) \implies \text{older}(x,y)$

Properties of Quantifiers



$\forall x \forall y$ is the same as $\forall y \forall x$

$\exists x \exists y$ is the same as $\exists y \exists x$

$\exists x \forall y$ is **not** the same as $\forall y \exists x$

$\exists x \forall y \text{ loves}(x,y)$ - "There is a person who loves everyone in the world"

بتتكمع شخص واحد

There exists a person x such that for every person y , x loves y .

بتتكمع كذا ع كل الناس

$\forall y \exists x \text{ loves}(x,y)$ - "Everyone in the world is loved by at least one person"

For every person y , there exists some person x who loves y .

Quantifier duality:

Each can be expressed using the other.

There are two important equivalences:

All students passed

$\forall x P(x) \equiv \neg \exists x \neg P(x)$ There is no student who failed

$\exists x P(x) \equiv \neg \forall x \neg P(x)$ مش كلهم مش بيحبو يعني شويه بيحبو

This means:

“For all” can be rewritten using “there exists” and negation.

“There exists” can be rewritten using “for all” and negation.

$\forall x \text{ likes}(x, \text{IceCream})$

$\neg \exists x \neg \text{likes}(x, \text{Icecream})$

$\exists x \text{ plays}(x, \text{Cricket})$

$\neg \forall x \neg \text{plays}(x, \text{Cricket})$

Self Test

- Every cat drinks milk.
- Virat loves Anushka
- Someone is Laughing
- Nobody loves George
- All men are mortal

$\forall x: \text{cat}(x) \rightarrow \text{drinks}(\text{milk})$

$\text{loves}(\text{virat}, \text{Anushka})$

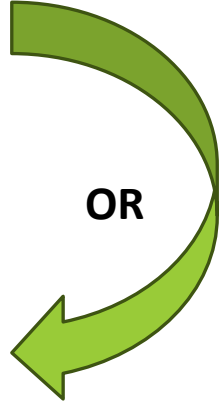
$\exists x: \text{laughing}(x)$

$\neg \exists x: \text{loves}(x, \text{George})$

$\forall x: \text{men}(x) \rightarrow \text{mortal}(x)$

$\forall x \neg \text{Loves}(x, \text{George})$

OR



طريقة تمثيل المعرفة باستخدام
Rules (قواعد)

2) PRODUCTION RULE

Production Rule is defined as a procedural representation of knowledge that links conditions to actions.

It allows domain experts to express their knowledge in the form of rules, making knowledge acquisition and maintenance easier.

For example, “if a customer automobile order includes the exterior appearance option, then add chrome trim and decals to the bill of materials.”

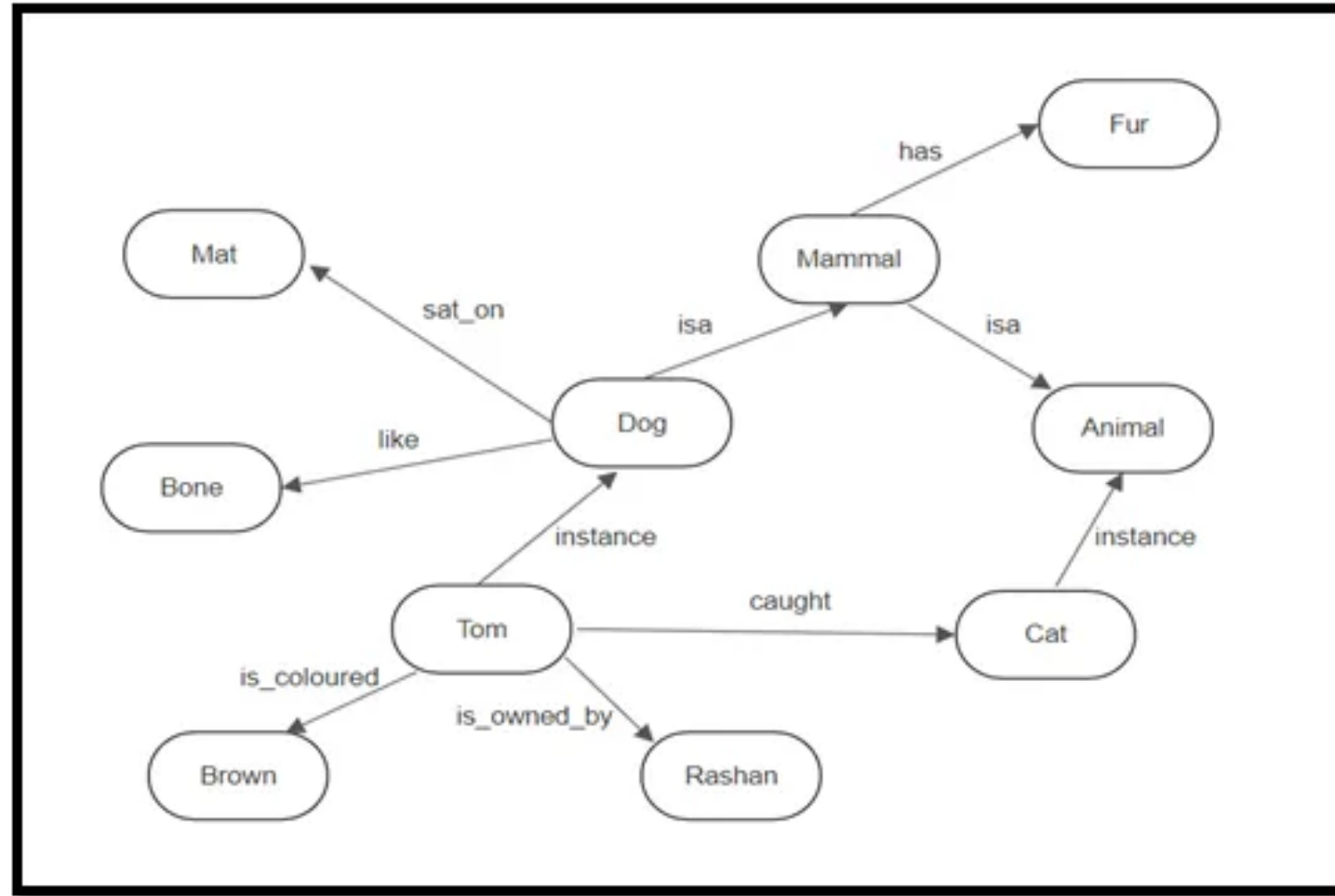
Semantic Network = Graph + Relationships

3) SEMANTIC NETWORK

These represent knowledge in the form of graphical network.

Example

- ☐ Tom is an instance of dog.
- ☐ Tom caught a cat
- ☐ Tom is owned by rashan.
- ☐ Tom is brown in colour.
- ☐ Dogs like bones.
- ☐ The dog sat on the mat.
- ☐ A dog is a mammal.
- ☐ A cat is an instance animal
- ☐ All mammals are animals.
- ☐ Mammals have fur.



Book Frame	
+-----+	
Slots:	
Title: "To Kill a Mockingbird"	
Author: "Harper Lee"	
Publication Year: 1960	
ISBN: "978-0-06-112008-4"	
Genre: "Fiction"	
+-----+	
Facets:	
Publication Year:	
- Type: Integer	
- Range: 1450 to current year	
ISBN:	
- Format: 13-digit number	
+-----+	
Default Values:	
Genre: "Unknown" (if not specified)	
+-----+	
Procedures:	
CheckAvailability: Method to check if the book	
is currently available in the library.	
UpdateRecord: Method to update the book's	
record when it is borrowed or returned.	
+-----+	

4) FRAME REPRESENTATION

Frames are record like structures that consist of a collection of slots or attributes and the corresponding slot value.

→ Slots have names and values called facets.